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The Silk Road Documentation

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1. Introduction

The Silk Road is an AI-driven mobile application designed to guide developers and software engineers through personalised learning paths. The app uses machine learning to assess user skills and preferences, generating tailored roadmaps that lead to mastery in specific tracks. It also features a robust community, gamification, and specialised content for students.

2. Key Features

2.1. Standard Questions

- **Description:** A set of standardised questions to assess the user's knowledge in problem-solving and computer science.
- **Impact:** Provides a foundation for generating personalised learning paths.
- **Technical Requirements:**
 - Mobile-friendly survey interface.
 - AI model integration for real-time analysis.

2.2. Generated Path or Roadmap

- **Description:** A personalised learning path based on the user's assessed skill level.
- **Impact:** Helps users navigate through the necessary learning stages effectively.
- **Technical Requirements:**
 - Dynamic content generation based on AI analysis.
 - Integration with mobile content management systems to curate resources.

2.3. Streaks (Challenge-Based Learning)

Description:

A streak-based feature where users are challenged with a series of tasks, broken down into three types of challenges, reinforcing learning and engagement through weekly goals. Streaks are maintained as long as users complete any of these challenges within the week.

Types of Challenges:

1. **General Challenges:**
 - Created and mentored by app admins.
 - These challenges focus on overall app-user engagement and provide a broad set of problems for users to tackle.
 - **Frequency:** Weekly.

2. Track-Specific Challenges:

- Created and mentored by track-specific mentors.
- These challenges are tailored to the individual learning tracks (e.g., Frontend, Backend, DevOps) that users are following.
- **Frequency:** Weekly.

3. Course-Specific Challenges:

- Created and mentored by course instructors.
- These challenges are based on the specific courses users are enrolled in and aim to deepen the understanding of course material.
- **Frequency:** Weekly.

2.4. Unique Developer Community

- **Description:** A specialized community for developers and software engineers to interact, share knowledge, and collaborate.
- **Impact:** Provides a platform for peer learning and networking.
- **Technical Requirements:**
 - In-app community channels.
 - Level-based access control and reward system.

2.5. Special Treatment for Students

- **Description:** A dedicated section offering resources and support tailored to students.
- **Impact:** Ensures that students receive focused guidance to complement their academic journey.
- **Technical Requirements:**
 - Student-specific content repository.
 - Mobile-friendly access to academic resources.

2.6. Gamification Elements

- **Description:** Incorporates game-like elements such as points, badges, and leaderboards to enhance user engagement.
- **Impact:** Increases motivation and user retention.
- **Technical Requirements:**
 - Reward and points system integration.
 - In-app leaderboard and achievement tracking.

2.7. Story-Driven Motivation

- **Description:** Each learning track includes a narrative to motivate users by connecting their learning goals with inspirational stories.
- **Impact:** Provides emotional motivation and context to the learning material.
- **Technical Requirements:**
 - Story content development.
 - Integration with track progression in the app.

2.8. Track-Specific Community Channels

- **Description:** Channels within the community for each learning track, with level-based access to ensure relevant discussions.
- **Impact:** Facilitates high-quality, focused discussions and learning support.
- **Technical Requirements:**
 - Channel management system.
 - User level assessment and verification.
 - In-app rewards for validated contributions.

2.9. Skills and Roadmap Role IQ

- **Description:** A structured assessment and learning path system that aligns with specific roles in software engineering.
- **Impact:** Guides users through a role-based learning experience, providing a clear progression path.
- **Technical Requirements:**
 - Role-based assessment tools.
 - Roadmap generation and progress tracking within the app.

2.10. Mentorship Option

Description: Provides users with the ability to request mentorship from experts within their chosen learning track.

- **Impact:** Offers personalized guidance, helping users overcome specific challenges and accelerate their learning.
- **Technical Requirements:**
 - **Mentor-Mentee Matching System:** Develop an algorithm that matches users with mentors based on skill level, goals, and track.
 - **In-App Communication Tools:** Implement features such as chat, video calls, or scheduled sessions to facilitate interaction between mentors and mentees.
 - **Progress Monitoring:** Allow mentors to track the progress of their mentees and provide feedback.
 - **Mentor Profiles and Ratings:** Create profiles for mentors, including their expertise and ratings from previous mentees.
 - **Integration with Learning Tracks:** Ensure that the mentorship option is seamlessly integrated into the user's current track, providing relevant advice and support.

This feature will not only enhance user engagement but also add value by offering personalized support, which can significantly improve learning outcomes.

3. User Flow and Processes

3.1. Splash Screen

- **Description:** The app's splash screen will display the logo and tagline, setting the tone for the user experience.
- **Technical Requirements:**
 - High-resolution logo.
 - Smooth transition to the onboarding screen.

3.2. Onboarding

- **Description:** A guided introduction that explains the app's features, how to use them, and the benefits for the user.
- **Steps:**
 - **Welcome Screen:** Brief introduction to the app's purpose.
 - **Permissions Request:** Request necessary permissions (e.g., notifications, data access).
 - **Feature Highlights:** Showcase key features with brief explanations.
 - **User Registration/Sign-In:** Allow users to create an account or sign in.
- **Technical Requirements:**
 - Smooth onboarding flow with minimal user friction.
 - Integration with social sign-in options.

3.3. Profile Setup

- **Description:** After registration, users can set up their profiles by adding information like their current skill level, goals, and areas of interest.
- **Steps:**
 - **Profile Customization:** Users choose their preferred learning track, set goals, and upload a profile picture.
 - **Skill Level Input:** Users can self-assess and input their skill levels if they have prior experience.
 - **Notifications Preferences:** Users can customize what types of notifications they want to receive (e.g., daily challenges, community updates).
- **Technical Requirements:**
 - Secure storage of user data.
 - Customizable notification settings.

3.4. Survey and Assessment

- **Description:** Users will complete a survey designed to assess their skill levels and preferences.
- **Steps:**
 - **Questionnaire Presentation:** Display the standard questions with a user-friendly interface.
 - **Response Collection:** Store responses securely for analysis.
 - **Skill Analysis:** Use AI to analyze responses and determine the user's level.
- **Technical Requirements:**

- Mobile-optimized survey interface.
- Real-time AI processing for skill assessment.

3.5. Dashboard (Home Screen)

- **Description:** The main hub where users can access their personalized roadmap, community, daily challenges, and other features.
- **Components:**
 - **Roadmap Overview:** Visual representation of the user's learning path.
 - **Daily Challenge:** Quick access to the daily problem-solving feature.
 - **Community Access:** Links to relevant community channels.
 - **Profile & Progress:** Display user profile, achievements, and streaks.
- **Technical Requirements:**
 - Mobile-friendly UI for the dashboard.
 - Real-time data synchronization.

3.6. Learning Path Execution

- **Description:** Users follow their personalized roadmap, engaging with curated content, tutorials, and exercises.
- **Steps:**
 - **Content Delivery:** Present users with the appropriate resources at each stage.
 - **Progress Tracking:** Monitor user's advancement through the roadmap.
 - **Adaptive Learning:** Adjust the roadmap based on user performance.
- **Technical Requirements:**
 - Content management system.
 - Real-time progress tracking and adaptive learning algorithms.

3.7. Community Interaction

- **Description:** Users interact within the community, participate in discussions, and seek help in specific channels.
- **Steps:**
 - **Channel Navigation:** Users select channels based on their learning track and level.
 - **Discussion Participation:** Users ask questions, share insights, and contribute to discussions.
 - **Reward System:** Users earn points and badges for helpful contributions.
- **Technical Requirements:**
 - In-app chat and discussion forums.
 - Moderation tools and reward distribution system.

3.8. Mentorship Interaction

- **Description:** Once a user is matched with a mentor, they can interact through in-app messaging or scheduled video calls.

- **Steps:**
 - **Messaging System:** Users and mentors can chat within the app to discuss goals, progress, and challenges.
 - **Session Scheduling:** Users can book video or voice calls with their mentors.
 - **Session Feedback:** After each session, users can provide feedback to improve the mentorship system.
- **Technical Requirements:**
 - Secure in-app messaging platform.
 - Integration with video calling APIs.
 - Feedback collection system.

3.9. Gamification and Rewards

- **Description:** Users engage with gamified elements like streaks, badges, and leaderboards to stay motivated.
- **Steps:**
 - **Streak Maintenance:** Users keep streaks by solving daily challenges.
 - **Achievement Unlocking:** Earn badges for milestones like completing a track or helping others in the community.
 - **Leaderboard Participation:** Users see where they stand in comparison to others.
- **Technical Requirements:**
 - Achievement tracking within the app.
 - Real-time leaderboard updates.

3.10. Student Special Section

- **Description:** A dedicated section offering resources and community support tailored for students.
- **Steps:**
 - **Student Dashboard:** Access to student-specific learning paths, resources, and community channels.
 - **Peer Collaboration:** Opportunities for students to collaborate on academic projects.
 - **Mentorship Opportunities:** Connect students with mentors from the developer community.
- **Technical Requirements:**
 - Student-specific content repository.
 - Mentor matching system integrated into the app.

3.11. Personalized Recommendations

- **Description:** Throughout the user journey, personalized content recommendations should be presented to guide them based on their progress and preferences.
- **Steps:**
 - **Resource Suggestion:** Recommend articles, videos, or courses based on the user's current position in their learning path.

- **Mentorship Suggestion:** Suggest mentors based on user progress and goals.
- **Technical Requirements:**
 - AI-driven recommendation engine.
 - Integration with the mentorship system.

3.12. Content Curation

- **Description:** Allow users to curate their learning resources, creating custom playlists or bookmarking important content for later review.
- **Steps:**
 - **Bookmarking:** Users can save resources like articles or videos to revisit later.
 - **Playlists:** Users can create custom playlists of content to follow a personalized study plan.
- **Technical Requirements:**
 - Bookmark and playlist management.
 - Syncing with the user profile.

3.13. Help and Support

- **Description:** Provide a help and support section within the app to assist users with common questions, technical issues, or feature usage.
- **Steps:**
 - **FAQ Section:** Users can access a list of frequently asked questions and solutions.
 - **Live Support:** Option to connect with customer support or community moderators for real-time assistance.
- **Technical Requirements:**
 - FAQ content management.
 - Chatbot or live chat integration.

3.14. User Feedback Loop

- **Description:** Collect continuous feedback from users to improve the app experience, including feature requests and bug reports.
- **Steps:**
 - **Feedback Prompt:** Periodically ask users to rate their experience or suggest improvements.
 - **Bug Reporting:** Provide an easy way for users to report issues they encounter.
 - **Survey Participation:** Offer occasional surveys to gather in-depth user opinions on new features or changes.
- **Technical Requirements:**
 - In-app feedback form.
 - Survey distribution tool.

3.15. Exit and Re-engagement

- **Description:** When users exit the app, they are encouraged to return through notifications and personalized content reminders.
- **Steps:**
 - **Session Continuation:** When users return to the app, start them exactly where they left off, whether it's a half-completed quiz, partially watched video, or ongoing streak.
 - **Re-engagement Incentives:** Offer special incentives or rewards to users who return after a period of inactivity, like bonus points or unlocking new content.
- **Technical Requirements:**
 - Push notification system.
 - Content reminders based on user activity and streaks.
 -

4. Technical Requirements

4.1. Frontend

- **Platform:** Mobile app development using Flutter and other tech .
- **Design:** Intuitive, mobile-first UI/UX design.
- **Authentication:** Integration with social sign-in options (Google, Facebook, etc.).

4.2. Backend

- **Database:** Secure, scalable cloud database for storing user data, survey responses, and community interactions.
- **AI Integration:** Machine learning models for real-time skill assessment and adaptive learning path generation.
- **Content Management:** CMS for handling dynamic content delivery and updates.

4.3. APIs and Integrations

- **Third-Party Services:** Integration with external APIs for content, notifications, and authentication.
- **Internal APIs:** Custom APIs for data processing, community interactions, and gamification features.

4.4. Security

- **Data Encryption:** End-to-end encryption for all user data.
- **Authentication Security:** Multi-factor authentication (MFA) for enhanced account security.
- **Compliance:** Ensure compliance with GDPR and other relevant data protection regulations.

5. Functional & Non-Functional Requirements.

5.1 Functional Requirements

1. Standard Questions for Skill Assessment

- **User Interface:**
 - Display a series of questions based on proven surveys.
 - Provide feedback on completion and assessment results.
- **Assessment Engine:**
 - Evaluate user responses to determine skill levels.
 - Generate a personalized learning path based on assessment results.

2. Generated Learning Path

- **Path Generation:**
 - Create a detailed learning path based on user skill levels and progress.
 - Include milestones and recommended resources.
- **User Tracking:**
 - Track user progress along the learning path.
 - Update the path dynamically as users advance or new skills are added.

3. Streaks (Daily Problem Solving)

- **Daily Problems:**
 - Present a new problem each day relevant to the user's learning track.
 - Track and display streak status in the user's profile.
- **Notifications:**
 - Notify users of their streak status and any rewards earned.

4. Developer Community

- **Community Channels:**
 - Create channels for specific learning tracks.
 - Allow users to post questions and receive answers from qualified peers.
- **Level-Based Access:**
 - Restrict responses to qualified users based on their track level.
 - Implement a reward system for active contributors.

5. Special Student Treatment

- **Student Section:**
 - Provide a dedicated section with resources tailored for students.
 - Include study guides, project ideas, and time management tips.
- **Support Features:**
 - Offer specialized support to help students balance learning with academic responsibilities.

6. Skill Gap Analysis

- **Skill Analysis Engine:**
 - Analyze the user's skills and compare them to industry standards.
 - Provide targeted recommendations for bridging skill gaps.
- **Progress Tracking:**
 - Allow users to track progress in closing skill gaps.

7. Personalized Career Path Prediction

- **Career Path Prediction:**
 - Analyze user skills, interests, and interactions to predict suitable career paths.
 - Present suggested career paths with explanations and steps to achieve them.

Additional Functional Requirements

8. Mentorship Matching System

- **Mentor Profiles:** Allow users to view profiles of available mentors, including their areas of expertise, track record, and user ratings.
- **Request Mentorship:** Enable users to request mentorship sessions, specifying their goals or challenges.
- **Session Scheduling:** Provide a built-in scheduling system to arrange mentorship sessions at mutually convenient times.
- **Mentorship Feedback:** Allow mentees to provide feedback on their mentorship sessions, helping to maintain high-quality guidance.

9. Resource Recommendation Engine

- **Dynamic Recommendations:** Based on user progress and performance, recommend articles, videos, and courses.
- **Resource Rating and Review:** Allow users to rate and review resources, helping others find the most effective materials.

10. Achievements and Badges

- **Achievement System:** Introduce badges for milestones like completing a certain number of problems, maintaining a streak, or reaching a new skill level.
- **Profile Display:** Display earned badges on the user profile to increase motivation and provide a sense of accomplishment.

5.2 Non-Functional Requirements

1. Standard Questions for Skill Assessment

- **Performance:**
 - Ensure quick load times for the survey.
 - The assessment engine must process responses efficiently.
- **Usability:**
 - The survey interface must be user-friendly and accessible across devices.

2. Generated Learning Path

- **Performance:**
 - The learning path should generate and update promptly.
- **Scalability:**
 - Support a growing number of users and learning paths.
- **Usability:**
 - The learning path must be easy to navigate and understand.

3. Streaks (Daily Problem Solving)

- **Performance:**
 - Ensure daily problems load quickly and correctly.
- **Usability:**
 - The streak-tracking system should be accurate and visually appealing.
- **Reliability:**
 - Ensure notifications are timely and reliable.

4. Developer Community

- **Scalability:**
 - Handle a large number of users and interactions within the community.
- **Security:**
 - Protect user data and ensure appropriate access control.
- **Usability:**
 - Community features should be intuitive and easy to use.

5. Special Student Treatment

- **Performance:**
 - Ensure the student section loads quickly and is responsive.
- **Usability:**
 - Resources and support must be relevant and accessible.
- **Scalability:**
 - Handle a large number of students and their diverse needs.

6. Skill Gap Analysis

- **Performance:**
 - The analysis should be performed quickly and efficiently.
- **Accuracy:**
 - Ensure the skill analysis and recommendations are accurate and based on current data.
- **Usability:**
 - Present analysis results in a clear and understandable format.

7. Personalized Career Path Prediction

- **Performance:**
 - The prediction engine should process user data efficiently.

- **Accuracy:**
 - Predictions must be based on accurate and up-to-date data.
- **Usability:**
 - The career path interface should be user-friendly and informative.

6. User Stories

User Story: Account Creation Using Email or Social Media

 **Plain Explanation** As a user, I want to create an account using my email or social media accounts, so I can access the app's features.

Related To

User Account Management

 **Details** Users should be able to sign up using either their email address or their social media accounts.

Acceptance Criteria

- Sign-up options for email and social media are available.
- Users receive confirmation of successful account creation.

 **Context and Background** Account creation is the entry point to accessing all app features. Providing multiple sign-up methods increases user convenience.

Dependencies

- Email service
- Social media integration

Non-Functional Requirements

- Secure registration methods
- Timely confirmation emails

Assumptions and Constraints

- Users must provide valid email or social media accounts.

Definition of Done

- Users can create an account using either email or social media and receive confirmation.

User Story: Secure Login

 **Plain Explanation** As a user, I want to log in securely using my credentials, so I can access my personalized track and progress.

Related To

User Account Management

 **Details** Users should be able to log in securely using their credentials.

Acceptance Criteria

- Login page is accessible and functional.
- Users can log in with correct credentials.
- User's session is secure and personalized.

 **Context and Background** Secure login is crucial for accessing personalized content and maintaining user trust.

Dependencies

- Authentication service
- User account management

Non-Functional Requirements

- Secure authentication methods
- Effective session management

Assumptions and Constraints

- Users must have valid credentials.

Definition of Done

- Users can log in securely and access their personalized content.
-

User Story: Password Reset

 **Plain Explanation** As a user, I want to reset my password if I forget it, so I can regain access to my account.

Related To

User Account Management

 **Details** Users should be able to reset their password through a secure process.

✓ Acceptance Criteria

- Password reset option is available.
- Users receive a reset link or code.
- Users can set a new password and access their account.

 **Context and Background** Password reset is essential for account recovery and maintaining continuous access.

Dependencies

- Email service
- Password reset service

Non-Functional Requirements

- Secure reset process
- User verification

Assumptions and Constraints

- Users must have access to the email or phone number associated with the account.

Definition of Done

- Users can reset their password and regain access to their account.
-

User Story: Profile Customization

 **Plain Explanation** As a user, I want to customize my profile by adding personal information, so the app can tailor content to my preferences.

Related To

Profile Customization

 **Details** Users should be able to add and update personal information in their profile.

✓ Acceptance Criteria

- Users can input and save personal details.
- Profile information is updated and used for personalized content.

 **Context and Background** Customizing profiles helps in tailoring the user experience by providing relevant content.

Dependencies

- User profile management system

Non-Functional Requirements

- Privacy settings
- Data validation

Assumptions and Constraints

- Personal information must be accurate.

Definition of Done

- User's profile is updated and used for tailoring content effectively.
-

User Story: Update Skill Levels

 **Plain Explanation** As a user, I want to update my skill levels in various areas of software engineering, so the app can adjust my learning track accordingly.

Related To

Profile Customization

 **Details** Users should be able to update their skill levels in different areas.

Acceptance Criteria

- Users can input and save skill levels.
- Learning track adjusts based on updated skills.

 **Context and Background** Updating skills ensures that users receive relevant learning paths aligned with their current knowledge.

Dependencies

- Skill assessment system

Non-Functional Requirements

- Data accuracy
- Real-time updates

Assumptions and Constraints

- Skill levels must be relevant and updated.

Definition of Done

- User's skill levels are updated and learning track adjusts accordingly.

User Story: Select Areas of Interest

 **Plain Explanation** As a user, I want to select my areas of interest in software engineering, so the app can suggest relevant tracks and resources.

Related To
Profile Customization

 **Details** Users should be able to choose areas of interest for personalized suggestions.

Acceptance Criteria

- Users can select and save interests.
- App suggests tracks and resources based on interests.

 **Context and Background** Selecting interests ensures that the app provides relevant content recommendations, enhancing the user experience.

Dependencies

- Interest management system

Non-Functional Requirements

- Personalized content
- Accurate suggestions

Assumptions and Constraints

- Interests must be relevant to the user's learning goals.

Definition of Done

- User receives relevant tracks and resources based on selected interests.
-

User Story: Complete a Survey for Skill Assessment

 **Plain Explanation** As a user, I want to take a survey or assessment to determine my current level of knowledge in specific areas, so I can be placed on the correct learning path.

Related To
Skill Assessment

 **Details** Users should be able to complete an assessment to determine their knowledge level.

✓ Acceptance Criteria

- Assessment is accessible and functional.
- Users receive results and are placed on the correct learning path.

 **Context and Background** Assessments are essential for accurately defining the right learning path for users.

Dependencies

- Assessment tool
- Skill evaluation algorithms

Non-Functional Requirements

- Accurate and fair assessments
- Data privacy

Assumptions and Constraints

- Assessments must be comprehensive and up-to-date.

Definition of Done

- User completes the assessment and is placed on the correct learning path.
-

User Story: Generate a Skill Profile

 **Plain Explanation** As a user, I want the app to generate a skill profile based on my survey results, so I can understand my strengths and areas for improvement.

Related To
Skill Assessment

 **Details** Users should receive a skill profile based on assessment results.

✓ Acceptance Criteria

- Skill profile is generated and accessible.
- Profile accurately reflects strengths and areas for improvement.

 **Context and Background** Skill profiles provide valuable insights into users' capabilities, helping them focus on areas that need improvement.

Dependencies

- Profile generation system
- Data analysis

Non-Functional Requirements

- Accurate profile generation
- Clear presentation

Assumptions and Constraints

- Profiles must be detailed and useful.

Definition of Done

- User receives a detailed skill profile that reflects their assessment results.
-

User Story: Receive a Personalized Learning Path

 **Plain Explanation** As a user, I want to receive a personalized learning path based on my skill assessment, so I can efficiently progress in my software engineering knowledge.

Related To

Personalized Learning Path

 **Details** Users should receive a learning path tailored to their skill assessment.

Acceptance Criteria

- Learning path is generated and accessible.
- Path is relevant to the user's skill level and learning goals.

 **Context and Background** Personalized learning paths enhance learning efficiency by focusing on relevant areas.

Dependencies

- Learning path generator
- Skill assessment results

Non-Functional Requirements

- Path clarity
- Progress tracking

Assumptions and Constraints

- Paths must be adaptable to user progress.

Definition of Done

- User receives and can follow a personalized learning path.

User Story: Manageable Learning Steps

 **Plain Explanation** As a user, I want to see my learning path broken down into manageable steps, so I can follow along without feeling overwhelmed.

Related To

Personalized Learning Path

 **Details** Users should see their learning path divided into manageable steps.

Acceptance Criteria

- Learning path is divided into clear, manageable steps.
- Users can track progress through each step.

 **Context and Background** Breaking down learning paths into steps helps users progress without feeling overwhelmed, making the learning process more approachable.

Dependencies

- Learning path visualization
- Progress tracking

Non-Functional Requirements

- Clear step division
- Progress indicators

Assumptions and Constraints

- Steps should be manageable and easy to follow.

Definition of Done

- User can follow a learning path divided into manageable steps.
-

User Story: Daily Coding Challenges

 **Plain Explanation** As a user, I want to receive daily coding challenges or problem-solving tasks, so I can improve my skills consistently.

Related To

Daily Challenges

 **Details** Users should receive daily challenges to practice coding and problem-solving.

✓ Acceptance Criteria

- Daily challenges are provided.
- Challenges are relevant to the user's skill level.
- Progress is tracked.

 **Context and Background** Daily challenges encourage consistent practice, leading to skill improvement over time.

Dependencies

- Challenge generator
- Progress tracker

Non-Functional Requirements

- Challenge relevance
- Clear instructions

Assumptions and Constraints

- Challenges should be challenging yet achievable.

Definition of Done

- User receives daily challenges and can track their progress.
-

User Story: Track Streaks

 **Plain Explanation** As a user, I want to track my streaks of daily challenges, so I can stay motivated to practice consistently.

Related To

Daily Challenges

 **Details** Users should be able to track their streaks when completing daily challenges.

✓ Acceptance Criteria

- Streaks are tracked and visible.
- Users receive notifications or rewards for maintaining streaks.

 **Context and Background** Tracking streaks provides motivation and rewards for consistent practice, encouraging long-term skill development.

Dependencies

- Streak tracker

- Notification system

Non-Functional Requirements

- Accurate streak tracking
- Reward system

Assumptions and Constraints

- Streak tracking should be reliable.

Definition of Done

- User can track their streaks and stay motivated to continue daily practice.
-

User Story: Interactive Community

 **Plain Explanation** As a user, I want to join an interactive community with discussions, so I can share knowledge and learn from others in my track.

Related To

Community Engagement

 **Details** Users should be able to join discussions and interact with the community.

Acceptance Criteria

- Users can join and participate in community discussions.
- Discussions are relevant to users' tracks.

 **Context and Background** An interactive community enhances learning through shared knowledge and support, creating a collaborative environment.

Dependencies

- Community management system

Non-Functional Requirements

- Community guidelines
- Moderation tools

Assumptions and Constraints

- Discussions should be relevant and respectful.

Definition of Done

- User can engage with the community and participate in meaningful discussions.
-

User Story: Gamification Elements

 **Plain Explanation** As a user, I want to see gamification elements such as badges, leaderboards, and rewards, so I can stay motivated and engaged in my learning journey.

Related To
Gamification

 **Details** Users should see and interact with gamification elements as they progress.

Acceptance Criteria

- Gamification elements like badges and leaderboards are implemented.
- Users receive rewards for achievements.

 **Context and Background** Gamification elements add an engaging, motivational layer to the learning experience, encouraging users to continue learning.

Dependencies

- Gamification engine
- Reward system

Non-Functional Requirements

- Fair and balanced gamification elements
- Clear criteria for rewards

Assumptions and Constraints

- Gamification should not distract from learning.

Definition of Done

- User sees and engages with gamification elements throughout their learning journey.
-

User Story: Mentorship Option

 **Plain Explanation** As a user, I want to request mentorship from someone proficient in my track, so I can receive guidance and support in my learning journey.

Related To
Mentorship

 **Details** Users should be able to request mentorship from experienced individuals in their track.

✓ **Acceptance Criteria**

- Mentorship request system is available.
- Users are matched with suitable mentors.

 **Context and Background** Mentorship provides personalized guidance, helping users overcome challenges and advance in their learning path.

Dependencies

- Mentorship management system

Non-Functional Requirements

- Mentor matching accuracy
- Privacy settings

Assumptions and Constraints

- Mentors should be knowledgeable and willing to help.

Definition of Done

- User can request mentorship and receive guidance from experienced individuals.
-

User Story: Student-Specific Features

 **Plain Explanation** As a student, I want to access features specifically designed for students, such as study groups and resources, so I can enhance my learning experience.

Related To
Student Engagement

 **Details** Students should have access to special features like study groups and resources tailored to their needs.

✓ **Acceptance Criteria**

- Student-specific features are available.
- Students can join study groups and access resources.

 **Context and Background** Providing student-specific features helps create a supportive learning environment, enhancing the overall learning experience.

Dependencies

- Student engagement tools

Non-Functional Requirements

- Resource relevance
- Study group management

Assumptions and Constraints

- Features should be useful and easy to access.

Definition of Done

- Student can access and benefit from student-specific features.
-

User Story: Student Space

 **Plain Explanation** As a student, I want to have a dedicated space where I can focus on my studies and engage with other students, so I can maximize my learning efficiency.

Related To
Student Engagement

 **Details** Students should have a dedicated space for focused study and interaction.

Acceptance Criteria

- Student space is available and accessible.
- Students can interact and study in a dedicated environment.

 **Context and Background** A dedicated space for students fosters a focused and collaborative learning environment, enhancing study efficiency.

Dependencies

- Student space management

Non-Functional Requirements

- Space customization
- Interactive tools

Assumptions and Constraints

- Space should be conducive to learning.

Definition of Done

- Student has access to a dedicated space for focused study and interaction.
-

User Story: Get Help in a Track Channel

 **Plain Explanation** As a user, I want to get help in a channel dedicated to my current track, so I can receive accurate and relevant assistance.

Related To
Community Engagement

 **Details** Users should be able to ask questions and get help in a track-specific channel.

Acceptance Criteria

- Track channels are available for user participation.
- Users can ask questions and receive relevant help.

 **Context and Background** Dedicated channels ensure that users receive help from peers or mentors familiar with the specific track, enhancing the quality of assistance.

Dependencies

- Channel management system

Non-Functional Requirements

- Channel moderation
- Accurate assistance

Assumptions and Constraints

- Channels should be focused and supportive.

Definition of Done

- User receives relevant help in the track-specific channel.
-

User Story: Contribute Effectively to Earn Rewards

 **Plain Explanation** As a user, I want to contribute effectively in discussions and help others in the community, so I can earn rewards and recognition.

Related To
Community Engagement

 **Details** Users should be rewarded for effective contributions in discussions and helping others.

✓ **Acceptance Criteria**

- Contribution tracking is in place.
- Users receive rewards for effective contributions.

 **Context and Background** Rewarding contributions encourages active participation and helps build a supportive community, enhancing the learning experience.

Dependencies

- Contribution tracking system
- Reward system

Non-Functional Requirements

- Fair contribution tracking
- Meaningful rewards

Assumptions and Constraints

- Contributions should be valuable and supportive.

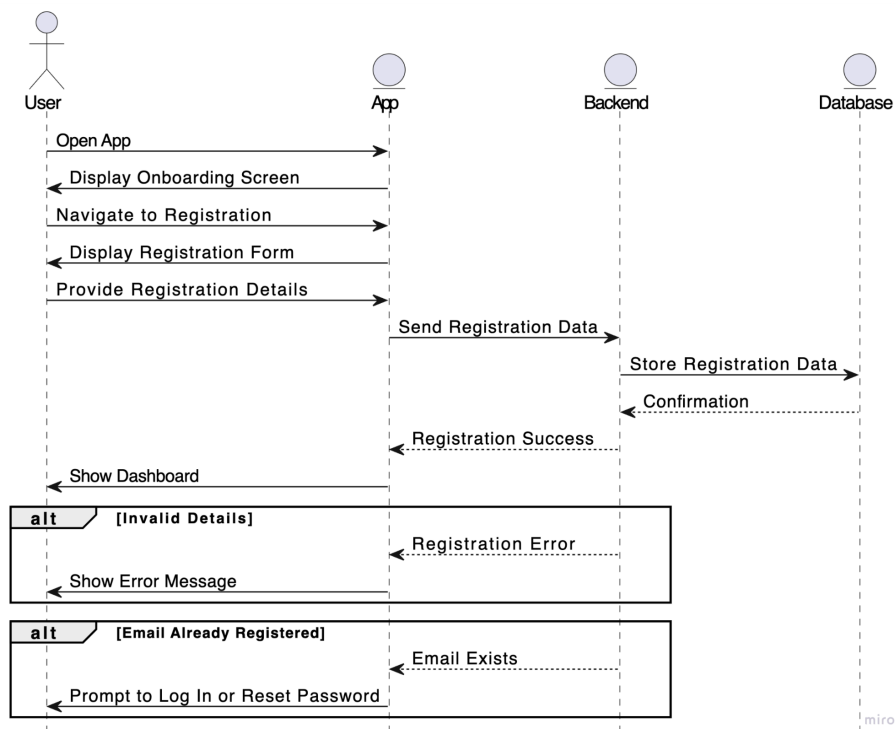
Definition of Done

- User is rewarded for effective contributions in the community.

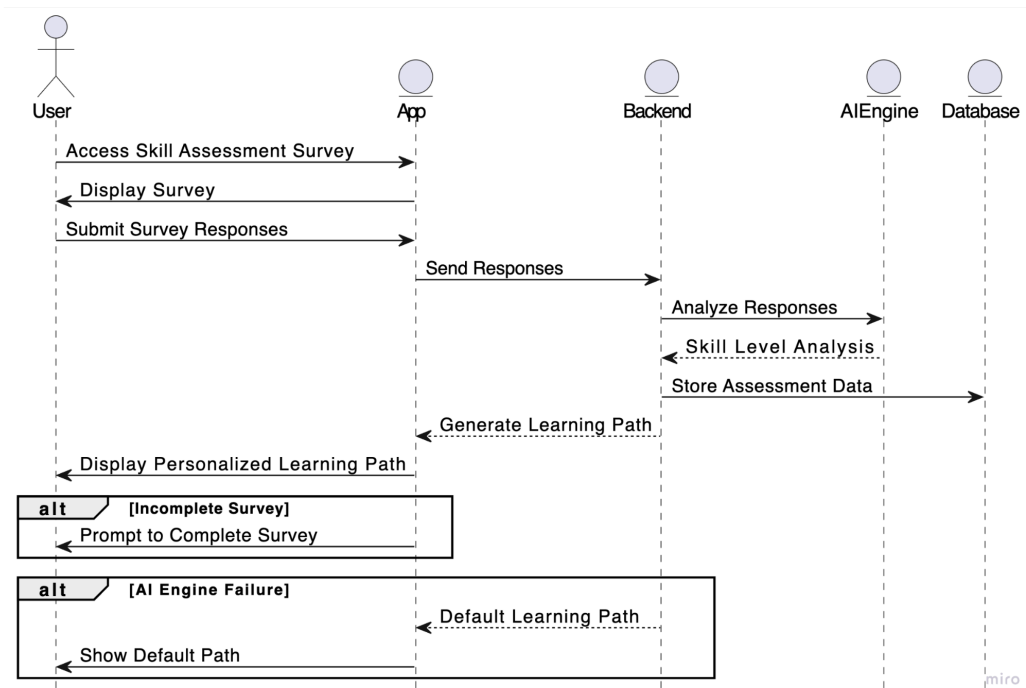
7. Diagrams and Flowcharts

7.1 Sequence Diagrams :

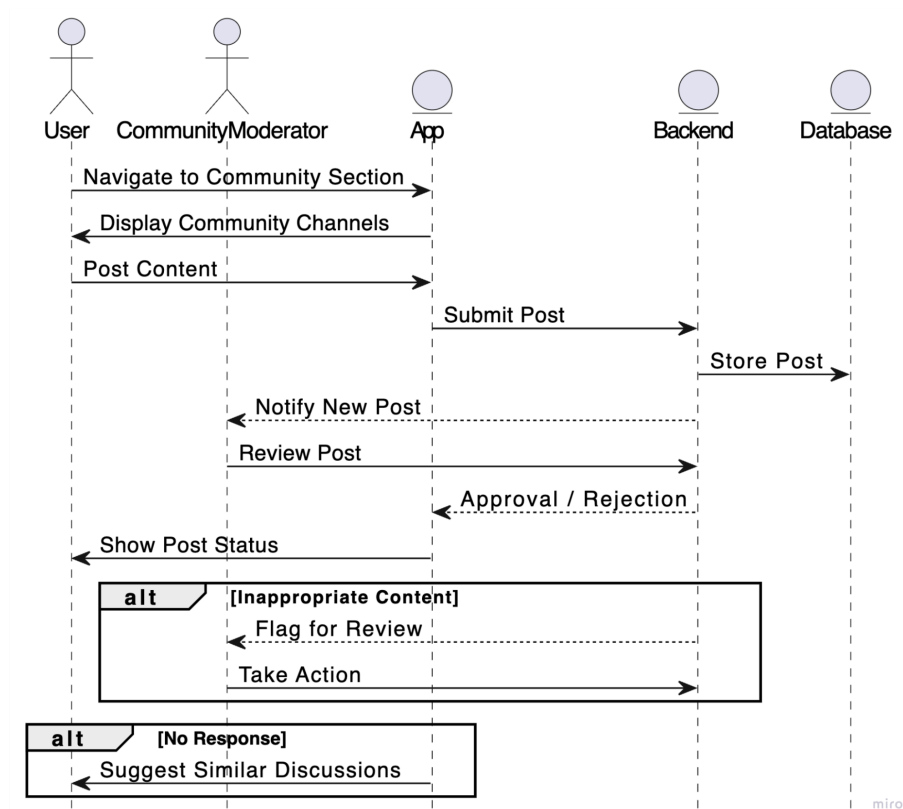
7.1.1 User Registration Process



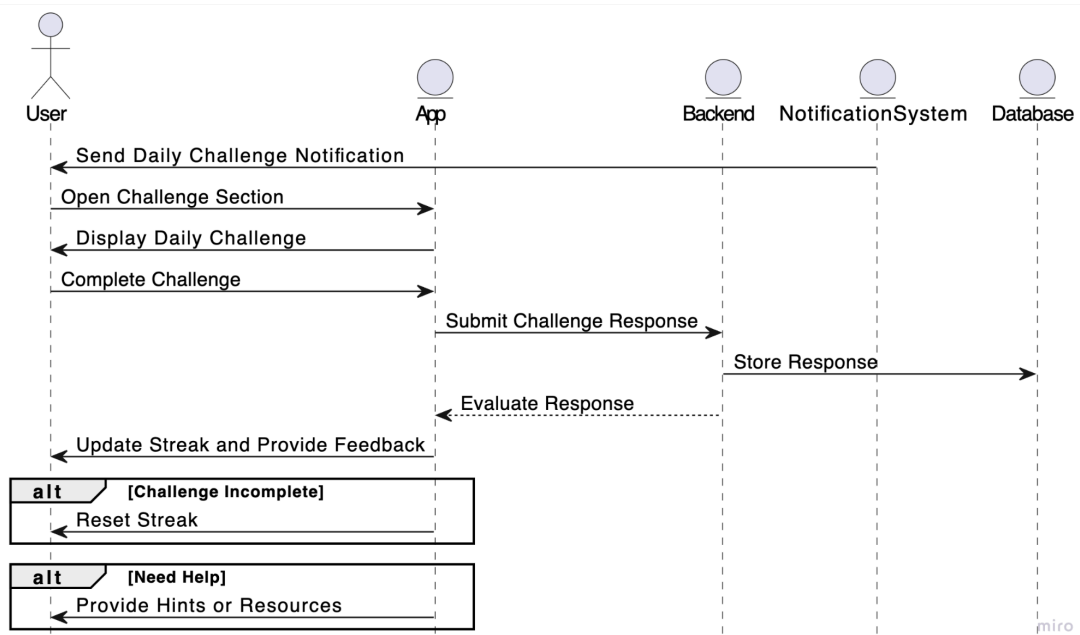
7.1.2 Skill Assessment and Learning Path Generation



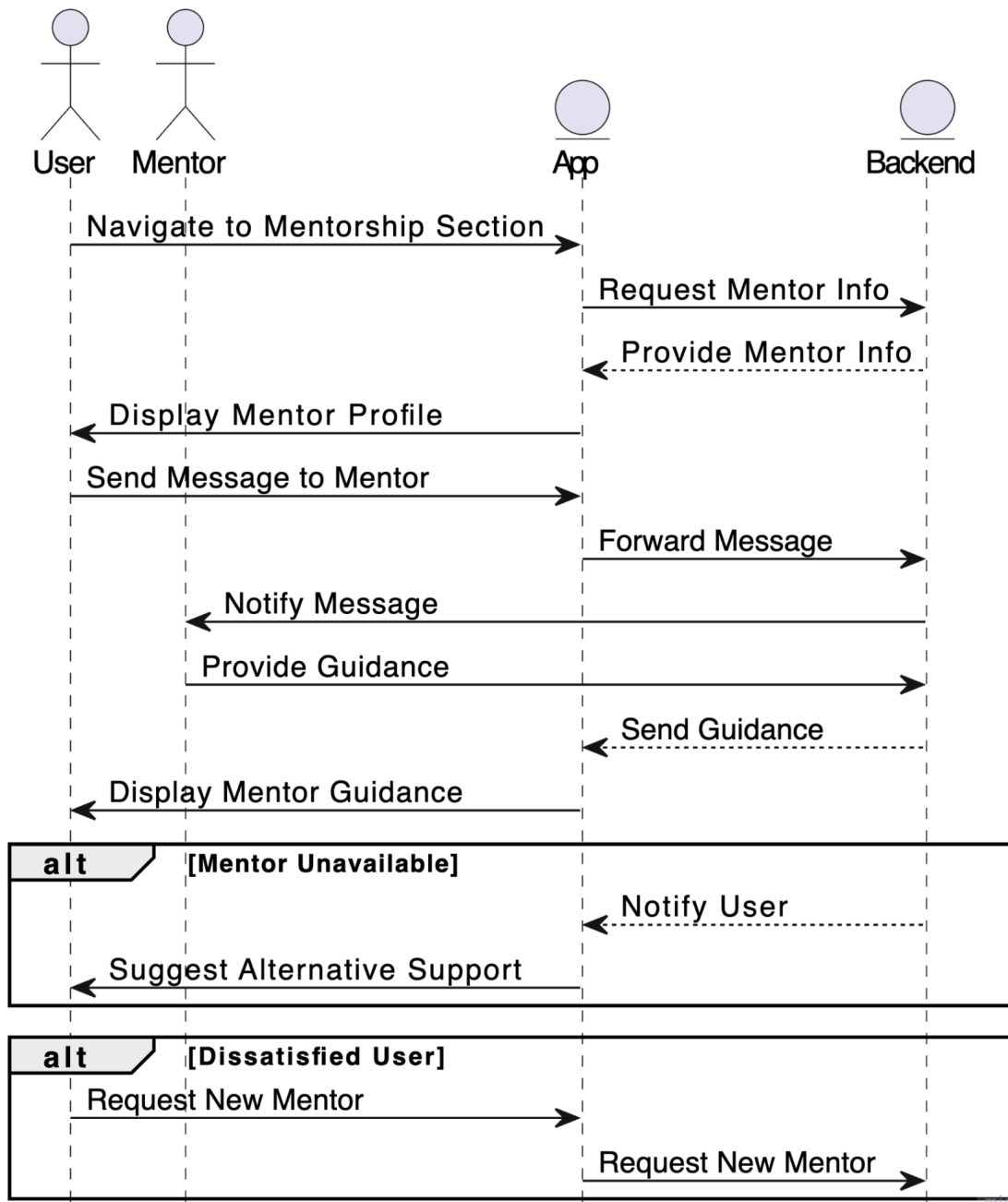
7.1.3 Community Engagement Interaction



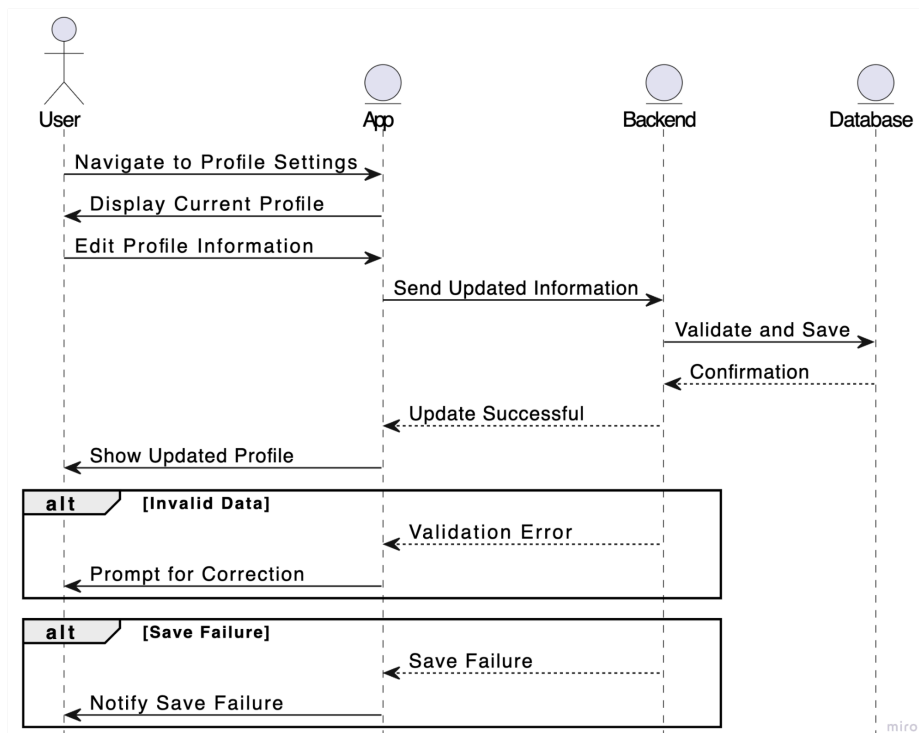
7.1.4 Daily Challenge Participation



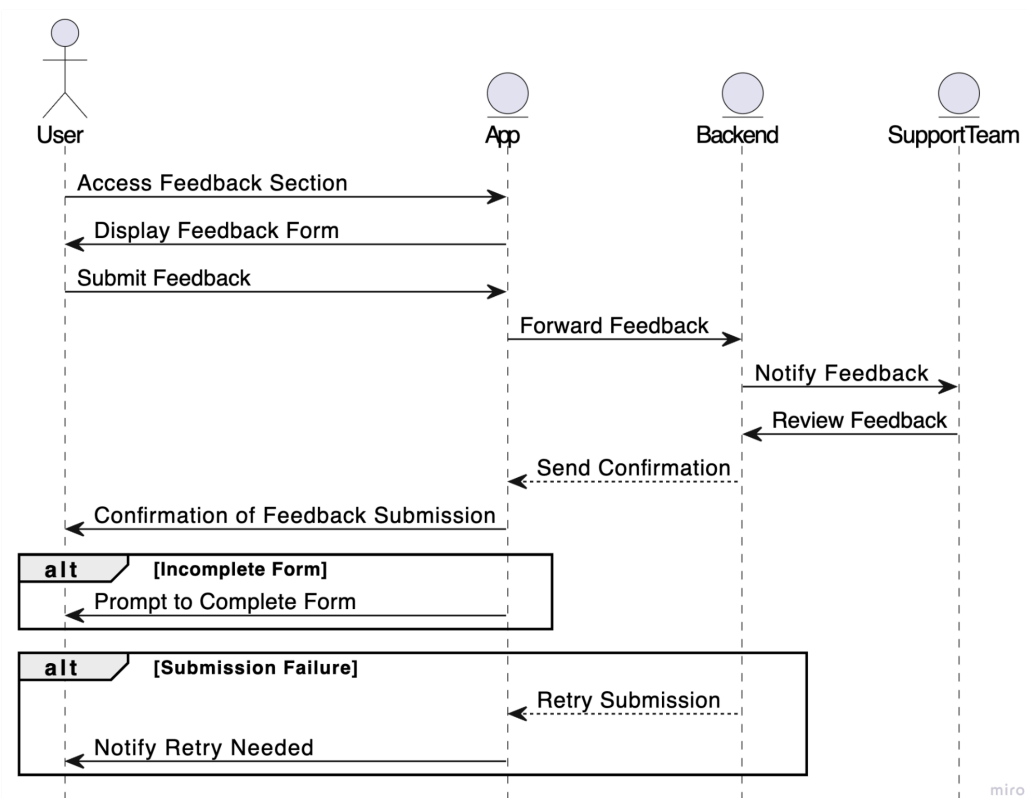
7.1.5 Mentorship Interaction



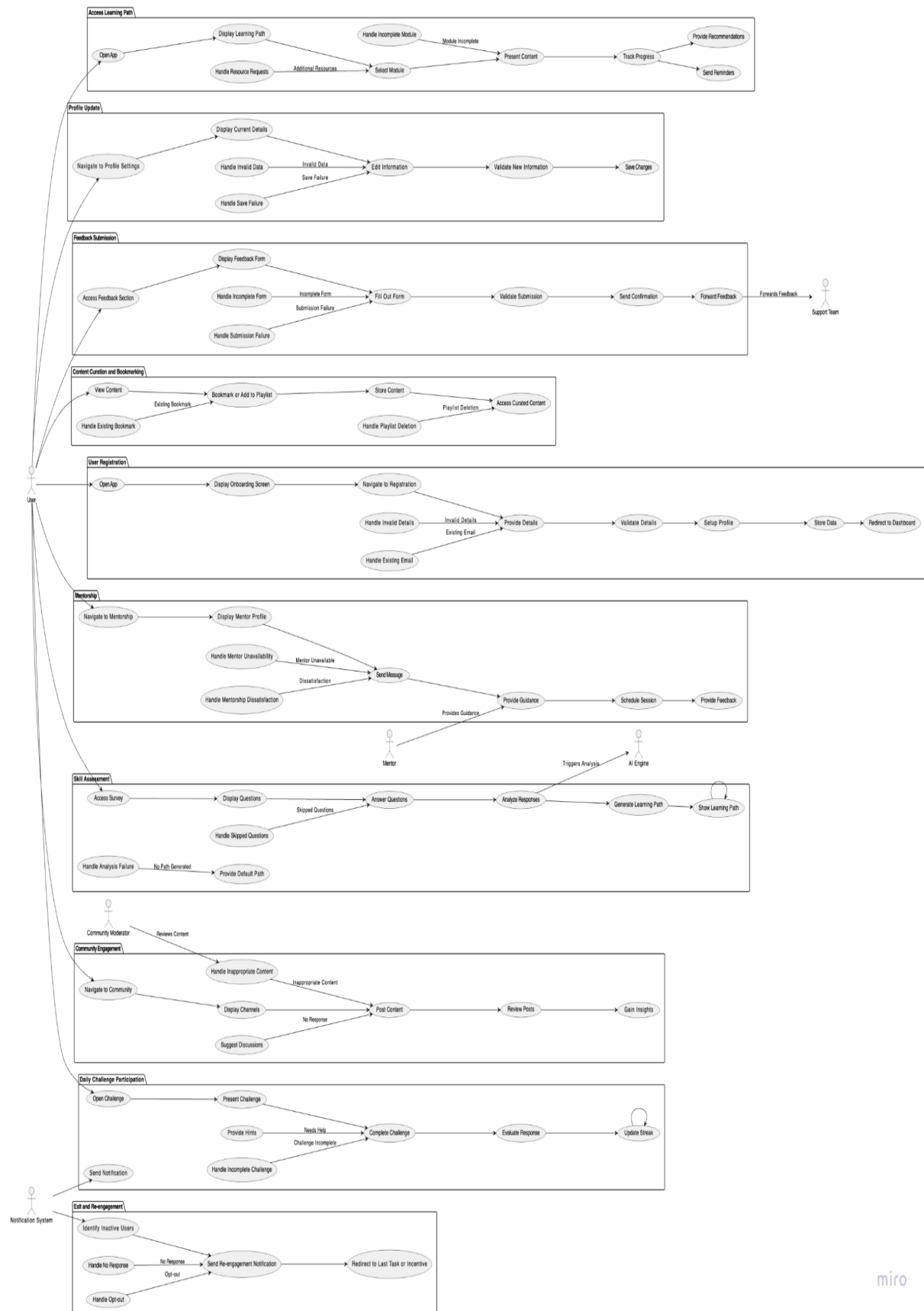
7.1.6 Profile Update Process



7.1.7 Feedback Submission Process



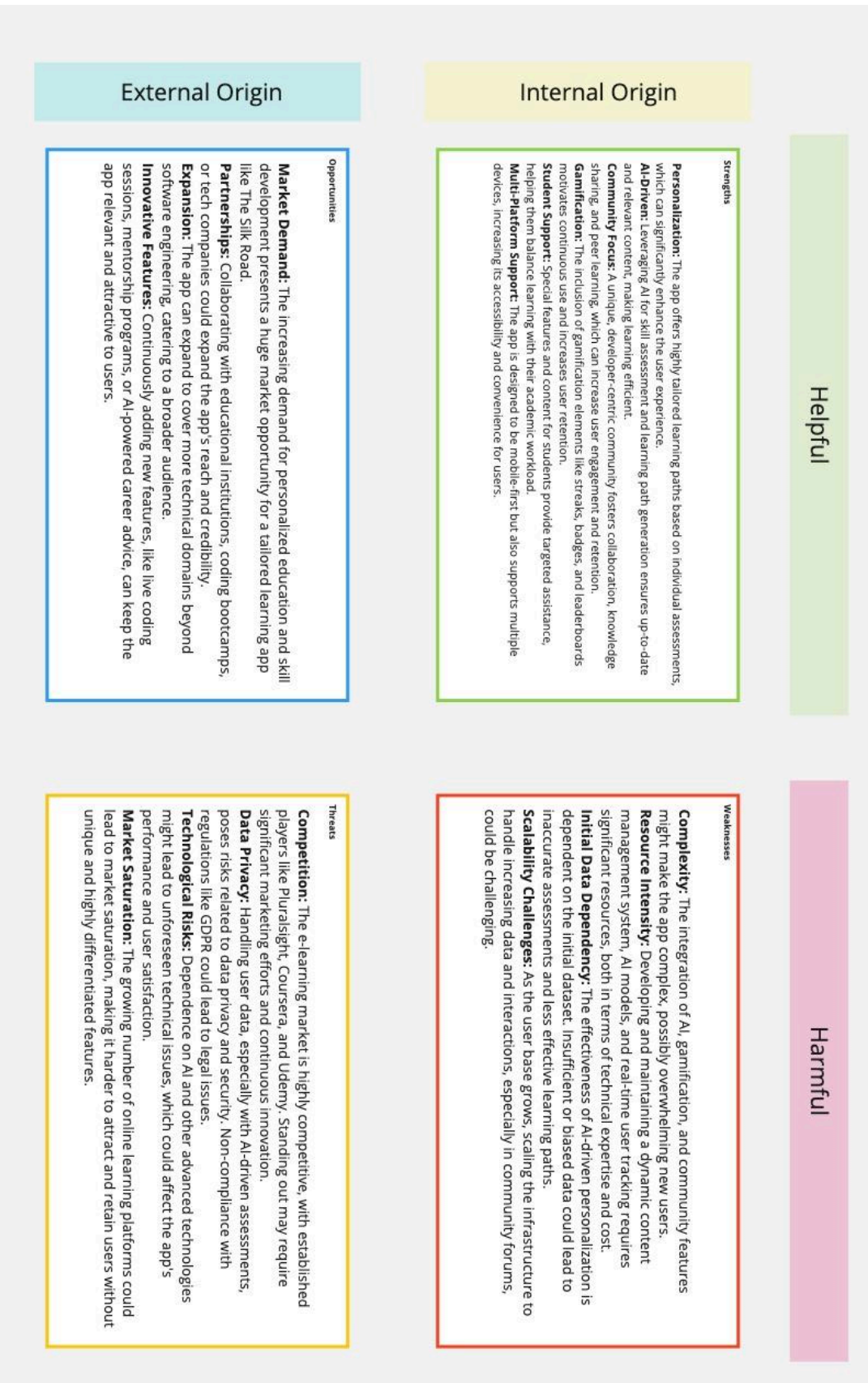
7.2 Use Case Diagram



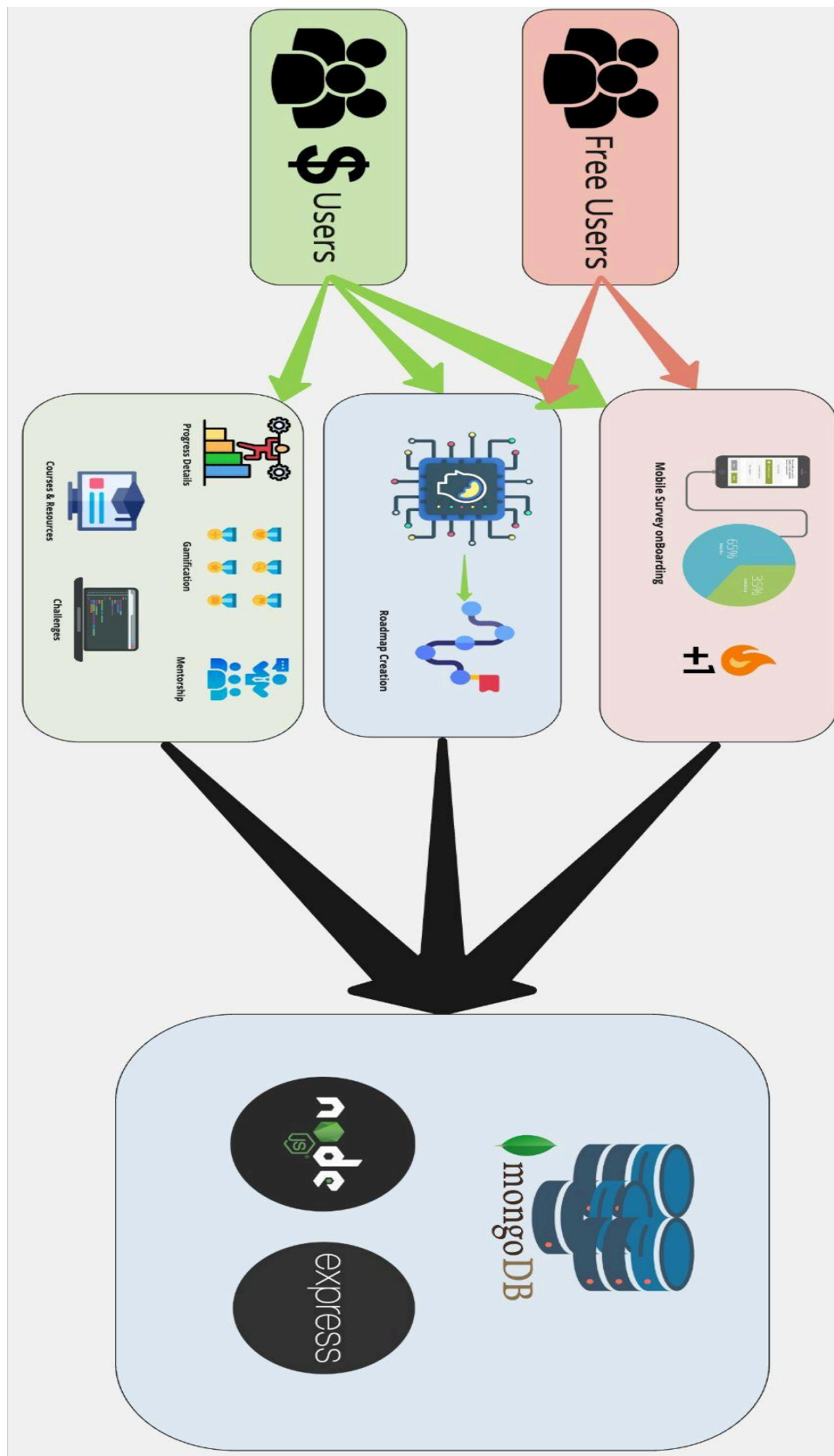
7.3 Class Diagram

On miro : https://miro.com/app/board/uXjVKkQtVlg=/?share_link_id=820029497829

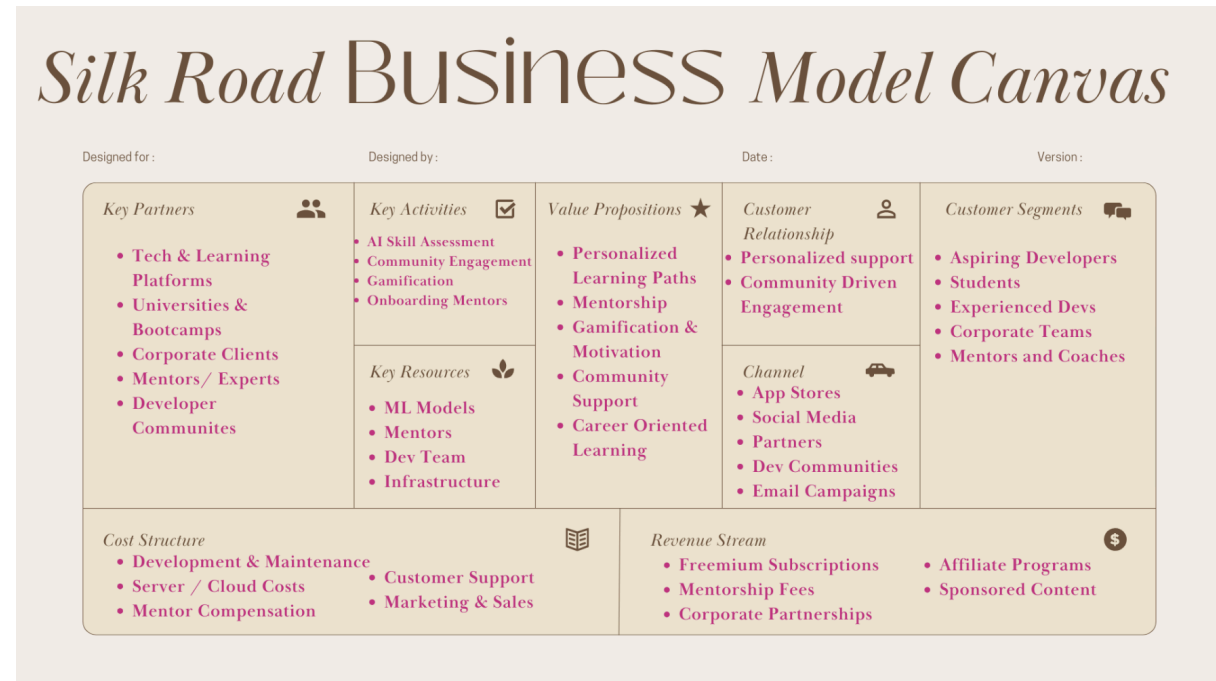
7.4 SWOT Analysis



7.5 Architecture Diagram



7.6 Business Model



Team Roles

Name	Role
EI Amir Mahmoud Mahmoud Mansour	Project Manager - Scrum Master
Ahmed Magdy Mohamed Elshwehy	Backend Developer
Zeyad Mohammed Mahmoud Salama	Backend Developer
Yousef SalahEldin Ebrahim Elgamily	AI Engineer
Dina Ibrahim Ragab	UI/UX Designer
Nourhan Ehab Al-ashquar	UI/UX Designer
Amira Shawki	Mobile Developer
Youssef Moustafa Ahmed	Mobile Developer

Timeline (Sprints)

